AutInsight Project

Complete technical, experimental, implementation, and research-paper handover

Purpose. This document is designed to be shared with project teammates and supplied to an LLM as grounded project context for drafting a research paper. It records the implemented system, exact available results, scope differences from the review presentation, limitations, paper structure, and rules against overclaiming.

Primary interpretation rule

The UCI target is exactly a threshold on the same ten item scores used as model inputs. UCI machine-learning metrics measure reproduction of that dataset label. They do not measure clinical autism detection.

Use this report with the source files listed in Section 15. If a number in this report conflicts with generated metrics JSON, treat the JSON and executable scripts as the implementation record and investigate the discrepancy before writing.

Prepared from the project workspace and the supplied Phase I Review II presentation. This report does not claim clinical validation, deployment, or external reproducibility beyond the recorded artifacts.

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Recommended use

Read Sections 1-10 before drafting. Draft methods and results only from Section 3-8 and verify them against Section 15 source files. Use Sections 11-14 to plan the manuscript and prepare answers. Do not import planned features or presentation-only numbers into the results section.

1. Executive summary and accurate project claim

AutInsight is a final-year research project consisting of (a) dataset acquisition and audit, (b) exploratory data analysis, (c) machine-learning experiments on two distinct pediatric ASD-related datasets, (d) SHAP explainability, and (e) a local caregiver-facing web prototype. It has a Next.js frontend and FastAPI backend. The primary saved inference artifact is an XGBoost pipeline trained on UCI dataset 419.

The main scientifically important result is a validity finding: UCI `result` equals the sum of A1-A10 for every row, and `class` is YES exactly when `result >= 7`. This makes the primary supervised target deterministic from the questionnaire input features. The direct rule reaches 100% on the held-out test data, and high model metrics should be described as reproducing the UCI class rule, not diagnosing ASD.

The separate Indian analysis uses the 2019 PLOS ONE S2 modified INDT-ASD cohort. Its target is the study's DSM-5 expert-consensus diagnosis, but the cohort is small and selected from suspected cases at a tertiary referral hospital. It is not pooled with UCI and does not validate the UCI model.

Recommended short description

AutInsight audits and models questionnaire-based pediatric ASD datasets, examines model behavior with SHAP, and presents a caregiver-oriented prototype. The work demonstrates a UCI label-construction problem and does not establish a clinically validated screening or diagnostic system.

Paper title suggestion

Auditing and Explaining Questionnaire-Derived Pediatric ASD Screening Labels: A Two-Cohort Machine-Learning Study and Caregiver-Facing Prototype

Claims the paper may make

- The project acquired/audited two public datasets and preserved their different target meanings.
- A deterministic audit found UCI target circularity: the target is derived from the same ten item inputs.
- Three model families were compared on one UCI holdout; two baseline models were evaluated separately on one Indian-cohort holdout.
- TreeSHAP was used for local primary XGBoost explanations and comparative Random Forest global/local research analysis.
- A local prototype organizes caregiver responses and presents model behavior with explicit limitations.

Claims the paper must not make

- The model diagnoses autism or achieves 97% clinical ASD accuracy.
- The model probability is a child's clinical probability or risk.
- SHAP reveals causes of autism or clinical importance of traits.
- The model is fair, calibrated, externally validated, or generalizable.
- The Indian cohort validates the UCI model.
- Counterfactuals, video analysis, or validated abstention are implemented.

2. Objectives, intended design, and architecture

Project objectives represented in the review materials include pediatric questionnaire-based screening research, careful preprocessing/evaluation, explanation of model predictions, subgroup fairness investigation, an accessible caregiver interface, and a future multimodal pathway. The concrete implementation covers data auditing, comparative tabular ML, partial subgroup analysis, SHAP, and a caregiver workflow. Some planned research components remain incomplete; see Section 10.

Layer	Implementation	Responsibility
Data/research	Python scripts under `scripts/`; pandas, NumPy, scikit-learn, XGBoost, SHAP	Acquire, validate, plot, train, evaluate, save results.
Frontend	Next.js 16.4.2, React 19.3.1, TypeScript; source under `frontend/src/app/`	Caregiver flow, summary, visual explanation, print actions.
API	FastAPI 0.115.12, Uvicorn 0.34.0, Pydantic 2 settings/schema	Validation, readiness, model metadata, inference, research JSON.
Model artifact	`models/primary.joblib`, manifest `models/artifact_manifest.json`	Saved UCI preprocessing/model pipeline and TreeSHAP explainer.
Research outputs	`experiments/reports/`, `experiments/results/`, `experiments/figures/`	Audits, experiment metrics, descriptive plots.

Online request path: Browser -> Next.js -> JSON POST to FastAPI `/predict` -> Pydantic validation -> saved preprocessing pipeline -> XGBoost `predict_proba` -> local TreeSHAP (when available) -> structured response -> frontend result page. The frontend does not perform the model inference itself.

Research endpoints are separate from prediction: `/research/metrics` and `/research/indian-metrics` return generated JSON. Operational routes are `/health`, `/ready`, and `/model`. The API root `/` has no page route and may return 404 by design.

The project has Docker/Compose and Vercel configuration, but deployment documentation says no cloud deployment or hosted URL exists. The local API can require OpenMP/libomp for XGBoost on macOS.

3. Data sources, cohorts, variables, and targets

3.1 UCI child screening data

Source: UCI Machine Learning Repository dataset 419, Autism Spectrum Disorder Screening Data for Children; DOI 10.24432/C5659W; CC BY 4.0 per project provenance documentation. Raw source: 292 records, 21 columns. Two exact duplicate rows are removed by the training script, leaving 290 modeling records. Raw label counts: 151 NO and 141 YES. The saved UCI holdout is 217 train / 73 test.

Field family	Fields/meaning	Model role
Questionnaire	A1_Score...A10_Score; binary item scores	Predictors; each item is also part of target generation.
Demographic/context	age (years), gender, ethnicity, jaundice, autism/family history, country_of_res, used_app_before, relation/respondent	Predictors in research models.
Derived total	result, range 0-10	Excluded predictor; equals the A1-A10 sum. Used to audit target.
Constant descriptor	age_desc	Excluded as constant.
Outcome	class, NO/YES	Target only; YES iff result >= 7.

Raw-data missingness: age 4/292; ethnicity 43/292; relation/respondent 43/292; no missingness in A1-A10, gender, jaundice, autism, country, prior-app, result, age_desc, or class in the EDA summary. Raw class counts are before deduplication.

The circularity cross-tab is decisive: NO rows occur only at scores 0-6 (counts 1, 6, 9, 21, 33, 41, 40); YES rows occur only at scores 7-10 (counts 44, 44, 32, 21). There is no overlap at a score. Training excludes the direct total field but retains the A1-A10 predictors.

3.2 Indian modified INDT-ASD study workbook

Source: PLOS ONE 2019 study supporting-information S2 workbook accessed via the project's Figshare download/provenance procedure; CC BY 4.0 is recorded in project docs. The workbook has 225 rows and 53 columns; no exact duplicates. Original target counts: 128 ASD (code 1) and 97 non-ASD (code 2). One 180-month record exceeds the paper's stated 14-year maximum (168 months); it is excluded from modeling, leaving 224 cases, 127 positive and 97 negative, split 168 train / 56 test.

Features (32): age in months; Gender; SES Kuppuswamy; Residence; and 28 modified INDT-ASD item fields: New 1a1-1a6, New1a7-New1a8, New1b1-New1b4, New1c1-New1c3, New2a1-New2a7, New2b, New2c, New2d1-New2d4. The 28 items are treated as categorical codes; age is numeric; gender/SES/residence are categorical.

Important missingness in raw workbook: SES Kuppuswamy 73; `newtoolseverity` 115; CARS Raw 1; each listed DP3 measurement/interpretation field 17. Many such measures are excluded as outcomes/proxies and are not model predictors. Data-quality report lists all 53 fields.

The Indian target is the study's DSM-5 gold-standard diagnosis, not the UCI target rule. The study cohort was children suspected of ASD at a North Indian tertiary referral hospital; this introduces selection/spectrum limits.

4. Audit, cleaning, and preprocessing

Common training/evaluation protocol

- UCI training removes exact duplicates before splitting. The Indian experiment removes one age outlier (>168 months) before splitting.
- Object columns are normalized by trimming whitespace/quotes and replacing sentinels such as `?`, empty string, `NA`, and `unknown` with missing values.
- UCI target labels are normalized to binary yes/no. Column aliases are normalized to shared names such as `family_history`, `country`, and `who_completed`.
- Pipelines are fit after a stratified 75/25 split with random_state=17. Training data alone fits imputation, scaling, encoding, and classifier.
- Numeric branch: SimpleImputer(median) -> StandardScaler. Categorical branch: SimpleImputer(most_frequent) -> OneHotEncoder(handle_unknown="ignore").
- UCI excludes `result`, constant `age_desc`, IDs, and constant fields; Indian excludes the ID, outcome summaries, severity and clinical proxy fields.

Precisely state the UCI validity issue

The direct total column `result` is excluded, so the obvious target column is not passed as an input. However, the target `class` is generated from the sum of the ten item scores, which remain inputs. This is a circular/score-derived target problem. Avoid the imprecise statement that model training accidentally includes `result` as a feature.

The train/test protocol is one holdout only. No k-fold, repeated, nested, site-based, or external validation appears in current training code. No confidence intervals for main model metrics, significance test, or hyperparameter optimization are implemented.

5. Models, hyperparameters, and software

Model	Exact project configuration	Data
Logistic Regression	max_iter=2000; class_weight="balanced"; random_state=17; sklearn defaults otherwise.	UCI and Indian
Random Forest	n_estimators=400; class_weight="balanced"; min_samples_leaf=2; random_state=17; sklearn defaults otherwise.	UCI and Indian
XGBoost	n_estimators=250; max_depth=3; learning_rate=0.03; subsample=0.85; colsample_bytree=0.85; eval_metric="logloss"; random_state=17; n_jobs=1.	UCI only; primary saved artifact

Logistic Regression models log-odds as a linear function of transformed predictors and is the linear baseline. Random Forest averages an ensemble of decision trees and can represent nonlinearities/interactions. XGBoost fits boosted trees sequentially, with shallow depth and shrinkage. All reported hard classifications use probability >=0.5. The default XGBoost classifier uses binary classification probability output; the artifact's model package version is 2.1.4.

XGBoost parameters were recorded in artifact manifest; the shared preprocessing pipeline includes imputation, scaling, and one-hot encoding. Scaling is useful to the linear model but is not typically required by tree splits. No hyperparameter optimization was conducted.

Runtime and package versions

Component	Recorded requirement/version
Python	>=3.11,<3.13; docs recommend Python 3.11 or 3.12.
scikit-learn	1.6.1 (manifest).
XGBoost	2.1.4 (manifest).
SHAP	requirements pin 0.46.0.
FastAPI / Uvicorn	0.115.12 / 0.34.0.
Frontend	Next.js 16.4.2, React/React DOM 19.3.1, TypeScript; Node >=20.19, Node 22 recommended.
Other pinned dependencies	pandas 2.2.3; numpy 2.2.3; scipy 1.15.2; joblib 1.4.2; matplotlib 3.10.0; seaborn 0.13.2; ucimlrepo 0.0.7; openpyxl 3.1.5.
Primary artifact	autinsight-0.1.0-uci419-seed17; 217 train, 73 test; seed 17; UCI dataset ID 419.

Artifact SHA-256: 4de746f481010e47472ce5d4d9fca01ce43b10bc44cc95992cef22abaa43a9bc. Dataset SHA-256 recorded in manifest: bd1fba8f42427bb6b5c5cd30f6b06a650693e3217658cd0dc19f8b9a60d9ec31.

6. Evaluation and exact model results

Metrics were computed on one stratified holdout (seed 17). Accuracy, balanced accuracy, F1, ROC AUC, and Brier score are reported. F1 uses the positive class; the decision threshold is 0.5. Brier is mean squared probability error on the dataset target and does not establish calibration.

Dataset/model	Accuracy	Balanced accuracy	F1	ROC AUC	Brier
UCI Logistic Regression	0.972603	0.973684	0.972222	1.000000	0.019219
UCI Random Forest	0.945205	0.943985	0.941176	0.988722	0.094480
UCI XGBoost	0.972603	0.972556	0.971429	0.997744	0.037459
UCI rule baseline	1.000000	1.000000	1.000000	not reported	not reported
UCI demographic-only LR	0.534247	0.533459	0.514286	0.583459	0.248386
Indian Logistic Regression	0.857143	0.848958	0.878788	0.945312	0.087559
Indian Random Forest	0.892857	0.880208	0.911765	0.967448	0.073109

The UCI score-rule baseline gets 1.0 accuracy/balanced accuracy/F1 by construction. UCI model performance must be interpreted in light of this. The demographic-only LR sensitivity model uses age, gender, ethnicity, jaundice, family history, country, previous app use, and respondent; its target remains the questionnaire-derived UCI class.

Indian metrics are from n=56 test records and should be reported separately. The Random Forest has higher point metrics than LR on this single split, but the experiment does not provide statistical testing or evidence of robust superiority.

7. Reliability tiers and subgroup outputs

7.1 P0-P3 engineering tiers

API confidence is max(p, 1-p), where p is the model positive-class probability. P0: >=0.90; P1: >=0.80; P2: >=0.70; P3: <0.70. P0/P1 count as accepted in the experiment calculation; P2/P3 are marked deferred in the API. These thresholds are provisional, uncalibrated, and not clinically justified.

UCI model	P0 count	P1 count	P2 count	P3 count	P0/P1 coverage	accepted errors
logistic_regression	56	9	2	6	0.8904	0
random_forest	13	14	14	32	0.3699	0
xgboost	42	13	5	13	0.7534	0

Observed selective risk was 0.0 in accepted cases for all three UCI models in this one test split. Accepted counts: LR 65/73 (coverage 89.04%); RF 27/73 (36.99%); XGBoost 55/73 (75.34%). These are small-split descriptive results, not validated confidence estimates. No calibration curve, isotonic/regression calibration, decision-cost threshold selection, entropy or Jensen-Shannon divergence is implemented.

Implementation mismatch to disclose: backend sets response `status="deferred"` for P2/P3, but the results UI does not enforce abstention and continues to render a positive/negative label. Therefore the project has provisional API tiers, not complete user-facing deferral.

7.2 UCI subgroup results: all groups

The table reports n, accuracy, sensitivity, and Wilson 95% interval for sensitivity. These group cells are descriptive; flag small n. Values are taken from generated `model_metrics.json`.

Model	Dimension / group	n	Accuracy	Sensitivity	Wilson 95% sensitivity CI
logistic regression	gender: f	21	0.952	1.000	[0.722, 1.000]
logistic regression	gender: m	52	0.981	1.000	[0.867, 1.000]
logistic regression	ethnicity: Asian	11	1.000	1.000	[0.510, 1.000]
logistic regression	ethnicity: Black	5	1.000	1.000	[0.342, 1.000]
logistic regression	ethnicity: Hispanic	1	1.000	1.000	[0.207, 1.000]
logistic regression	ethnicity: Latino	2	1.000	1.000	[0.207, 1.000]
logistic regression	ethnicity: Middle Eastern	7	1.000	1.000	[0.438, 1.000]
logistic regression	ethnicity: Missing	13	0.923	1.000	[0.510, 1.000]
logistic regression	ethnicity: Others	2	1.000	NA	NA
logistic regression	ethnicity: South Asian	6	1.000	1.000	[0.438, 1.000]
logistic regression	ethnicity: White-European	26	0.962	1.000	[0.816, 1.000]
random forest	gender: f	21	0.952	0.900	[0.596, 0.982]
random forest	gender: m	52	0.942	0.920	[0.750, 0.978]
random forest	ethnicity: Asian	11	1.000	1.000	[0.510, 1.000]
random forest	ethnicity: Black	5	1.000	1.000	[0.342, 1.000]
random forest	ethnicity: Hispanic	1	1.000	1.000	[0.207, 1.000]
random forest	ethnicity: Latino	2	1.000	1.000	[0.207, 1.000]
random forest	ethnicity: Middle Eastern	7	0.714	0.333	[0.061, 0.792]
random forest	ethnicity: Missing	13	0.846	0.750	[0.301, 0.954]
random forest	ethnicity: Others	2	1.000	NA	NA
random forest	ethnicity: South Asian	6	1.000	1.000	[0.438, 1.000]
random forest	ethnicity: White-European	26	1.000	1.000	[0.816, 1.000]
xgboost	gender: f	21	0.952	0.900	[0.596, 0.982]
xgboost	gender: m	52	0.981	1.000	[0.867, 1.000]
xgboost	ethnicity: Asian	11	1.000	1.000	[0.510, 1.000]
xgboost	ethnicity: Black	5	1.000	1.000	[0.342, 1.000]
xgboost	ethnicity: Hispanic	1	1.000	1.000	[0.207, 1.000]
xgboost	ethnicity: Latino	2	1.000	1.000	[0.207, 1.000]
xgboost	ethnicity: Middle Eastern	7	0.857	0.667	[0.208, 0.939]
xgboost	ethnicity: Missing	13	0.923	1.000	[0.510, 1.000]

Model	Dimension / group	n	Accuracy	Sensitivity	Wilson 95% sensitivity CI
xgboost	ethnicity: Others	2	1.000	NA	NA
xgboost	ethnicity: South Asian	6	1.000	1.000	[0.438, 1.000]
xgboost	ethnicity: White-European	26	1.000	1.000	[0.816, 1.000]

7.3 Indian gender subgroup results

Model	Source gender code	n	Sensitivity	Wilson 95% sensitivity CI
logistic regression	1	33	0.833	[0.608, 0.942]
logistic regression	2	23	1.000	[0.785, 1.000]
random forest	1	33	0.944	[0.742, 0.990]
random forest	2	23	1.000	[0.785, 1.000]

No subgroup fairness guarantee is supported. UCI ethnicity groups include n=1, n=2, n=5, n=6, and n=7 cells; Wilson intervals are wide. Indian subgroup values are based on n=33 and n=23 records by source gender code. Do not infer fairness from high point estimates.

8. Explainability: algorithms, outputs, checks, and gaps

8.1 SHAP concept

SHAP means SHapley Additive exPlanations. For an input x, the additive explanation is model_output(x) = baseline value + sum of feature contributions. Each contribution represents a model attribution relative to a chosen background/reference distribution under SHAP's feature-coalition assumptions. Positive and negative signs move the explained output in opposite directions. They do not establish causality.

TreeSHAP computes Shapley-style values efficiently for tree ensembles. It is used because XGBoost and Random Forest are tree models.

8.2 Primary XGBoost local API explanation

- Model artifact stores the fitted XGB pipeline, feature names, TreeExplainer, and metadata.
- At request, runtime constructs one-row DataFrame in expected feature order; calls the fitted preprocessor; computes explainer.shap_values on transformed input; extracts positive-class values if output has class dimension; sorts absolute values descending; returns top 10.
- API schema labels explanation scale `raw_model_score`. The frontend describes direction as toward UCI YES/NO. Bar width is scaled to the maximum absolute contribution within that case.
- The explanation page is not a conventional TreeSHAP waterfall or beeswarm. Feature values such as encoded category dummy variables may be displayed; mapped question labels are human-readable when recognized.
- XGBoost SHAP additivity was not independently verified. SHAP can fail at runtime; service logs exception and returns an empty explanation list.

8.3 Comparative Random Forest research SHAP

Training analysis transforms the UCI test features, calculates RandomForest TreeExplainer values, averages absolute positive-class values across held-out cases, ranks the top 10 globally, and saves one local example (test index 0). RF baseline plus local SHAP contributions was compared with RF positive-class probability; reported additivity error = 1.887379141862766e-15.

RF global top ten mean absolute SHAP values

Rank	Transformed feature	Mean |SHAP|
1	numeric__a4_score	0.09547387
2	numeric__a9_score	0.07610931
3	numeric__a10_score	0.06112183
4	numeric__a5_score	0.05796733
5	numeric__a8_score	0.05697540
6	numeric__a1_score	0.05550532
7	numeric__a7_score	0.03907448
8	numeric__a6_score	0.03899657
9	numeric__a3_score	0.03270961
10	numeric__a2_score	0.02696514

RF held-out example local contributions

Rank	Transformed feature	Signed contribution
1	numeric__a4_score	-0.12837446
2	numeric__a9_score	+0.08672685
3	numeric__a1_score	-0.07876213
4	numeric__a8_score	+0.06860773
5	numeric__a10_score	+0.04367267
6	numeric__a5_score	+0.04008724
7	numeric__a6_score	+0.03607582
8	numeric__a7_score	+0.03282509
9	numeric__a2_score	+0.02490565
10	numeric__a3_score	+0.02246327

The leading global RF features are all questionnaire items. In view of the target construction, this reflects the model's recovery of a rule defined from those same items. Do not call it a clinical ranking of autism traits.

8.4 Planned versus implemented explainability

- Implemented: local XGBoost TreeSHAP contributions in API; global mean absolute SHAP and one local example for comparative RF; one RF additivity check.
- Not implemented: XGBoost additivity verification; SHAP stability, perturbation fidelity, or user studies; LIME comparison; counterfactual algorithm; SHAP bar/beeswarm/waterfall plot suite as specified in review slides.
- Counterfactual field exists as optional nullable response property but no generator populates it. Domain-approved mutable-feature constraints are a prerequisite.
- No explanation is causal, a diagnosis, or evidence of clinical importance.

9. Caregiver application, API, privacy, deployment

9.1 Caregiver flow

Step	Behavior
Home	Introduces tool, its limits, and entry to screening.
Screening step 1	Age in years gate 4-11; ten adapted AQ-10 Child-Version-informed paraphrases; four choices: definitely agree, slightly agree, slightly disagree, definitely disagree.
Scoring	Four-choice responses collapse to binary. Agree-keyed items 1, 5, 7, 10; disagree-keyed 2, 3, 4, 6, 8, 9. Item order preserved for UCI model mapping.
Screening step 2	Duration, settings, areas for discussion, strengths, regression, previous professional support. Context observations are not scored by model.
API request	Age and 10 binary answers sent to `/predict`; gender/jaundice/family history hardcoded unknown; other fields rely on schema defaults (ethnicity/respondent/country Unknown; previous app no).
Results	Explains UCI source-data class, score, local feature contributions, context, practical next steps, disclaimer, print.
Understand page	Shows case-specific signed contribution bars, static dataset research plots, and explanations of methods.

The item wording is paraphrased, and the online flow has not been validated as a clinical questionnaire. The full second-stage caregiver context remains in tab sessionStorage and is not sent to model or stored in backend database. Results also use sessionStorage and disappear with tab/session reset.

9.2 API contract

Input schema: `age` float 4-11; `gender` Literal m/f/unknown; `jaundice` yes/no/unknown; `family_asd` yes/no/unknown; optional ethnicity, respondent, country, used_app_before defaults; exactly ten binary integer `questions`. Invalid values yield HTTP 422. Output fields include status, screening_label, probability, reliability_tier, message, explanation list, explanation_scale, limitations, model_version, optional counterfactual.

Inference row includes A1-A10, age, gender, ethnicity, jaundice, family_history, country, used_app_before, who_completed. The current UI supplies little of the non-question demographic information and uses defaults. One-hot encoder ignores categories never seen in training; inference defaults can therefore matter. Document this behavior.

9.3 Privacy and security status

No authentication, rate limiting, formal privacy impact assessment, retention controls, or clinical governance. UI asks users not to enter identifiers. Request bodies are not intentionally persisted by the API; hosting logs may include operational metadata. Raw data is git-ignored. Joblib artifacts must be trusted because pickle-based deserialization is unsafe for untrusted artifacts. CORS allowlists origins. No production deployment exists.

9.4 Runtime and deployment

Python requirement >=3.11,<3.13; Node >=20.19, recommended 22. `requirements.txt` pins API/model dependencies. On macOS, XGBoost needs compatible OpenMP (`libomp`). Vercel root is `frontend`; FastAPI container is configured in `backend/Dockerfile`; Docker Compose is available. `NEXT_PUBLIC_API_URL` selects API origin; `ALLOWED_ORIGINS` selects frontend CORS origins. Local ports 3000 and 8000 are used.

10. Review presentation: proposed scope versus actual state

This comparison treats the Phase I Review II slide deck as the intended project proposal. Presentation-only planned values must not be presented as outputs generated by current code.

Presentation idea	Current implementation / honest status
Pediatric, age-aware screening patterns	Caregiver flow accepts ages 4-11 and uses UCI child questionnaire. No cross-age child/adolescent/adult cohort analysis.
Indian/local population relevance	Indian S2 cohort is separately audited and modeled. It is not the deployed model; no local-population validation is established.
Streamlit dashboard	Frontend is Next.js/React; backend is FastAPI.
LR, SVM, RF, XGBoost / neural candidate	UCI: LR, RF, XGBoost. Indian: LR, RF. No SVM or neural model in current scripts.
Five-fold cross-validation; calibration	One stratified 75/25 holdout. No CV; no calibration model or curve.
JSD/entropy reliability selection	Only uncalibrated P0-P3 max-probability thresholds. No JSD or entropy. Frontend does not enforce deferred status.
SHAP bar, beeswarm, waterfall	Local XGBoost signed contribution bars; comparative RF global and local SHAP values in JSON. The full plot suite is not implemented.
Counterfactual recourse	Deferred; no generation code.
Fidelity, sparsity, proximity, validity, plausibility, stability	Only one RF additivity reconstruction check. Other explanation and counterfactual quality metrics not run.
Subgroup fairness	Descriptive UCI gender/ethnicity and Indian gender metrics with intervals; insufficient for a fairness claim.
Video/OpenPose/CNN-LSTM	Future phase only; not implemented.
Remove doctor from screening loop	Prototype gives professional next-step guidance and is not a substitute for a clinician.
Presentation Slide 12 neural/reliability numbers	99.31% macro-F1, 99.32% micro-F1, p=.6127, 97.95% coverage, JSD/entropy coverage values are not produced by current repository scripts/artifacts. Do not report as project experiment results without separate reproducible evidence.

Related review slides describe literature-paper methods and findings (for example, base-paper results). Those are claims about cited studies, not results of AutInsight. Verify original references before repeating them.

11. Research paper outline and approximate word allocation

Target approximately 5,500-6,500 words excluding references, tables, and figure captions, subject to institutional format.

Section	Words	Required content
Title, abstract, keywords	200-250 abstract	Objective, cohorts/targets, models, split, headline results, UCI circularity, limits. 4-6 keywords.
1 Introduction	600-750	Need and context; screen vs diagnosis; validity/explainability gaps; aim/RQs/contribution.
2 Related work	700-900	Questionnaire ML, age transfer, clinical target validation, calibration/deferral, SHAP/XAI, Indian study. Verify papers.
3 Data and target definitions	600-750	Source, license, cohorts, variables, counts, inclusion/exclusion, targets and source population.
4 Audit/preprocessing	400-550	Quality checks, circularity, duplicates, missingness, exclusions, pipeline and train-only fitting.
5 Models	500-650	Exact LR/RF/XGB algorithms, hyperparameters, pipeline, seed, cutoff and artifact. Indian uses LR/RF only.
6 Evaluation protocol	450-600	Split sizes, metrics, subgroup methods, no CV/external validation.
7 Explainability/reliability	500-700	TreeSHAP concept, local/global distinction, model scope, additivity result, tier thresholds and limits.
8 Application prototype	300-450	Caregiver flow, API, fixed default fields, sessionStorage, disclaimer.
9 Results	550-750	Separate tables per cohort; direct UCI rule and demographic-only result; subgroup/SHAP findings.
10 Discussion	600-800	Interpret target circularity; separate Indian interpretation; restrained XAI and fairness conclusions.
11 Limitations/ethics/future work	500-700	Holdout/sample limits, validity, calibration, subgroup, adapted tool, privacy, no video/counterfactual.
12 Conclusion	150-250	Restate audit, exploratory modeling, XAI, prototype, no clinical validation.

Suggested RQs

- RQ1: How well do LR, RF, and XGBoost reproduce UCI labels in a stratified holdout?
- RQ2: What does deterministic audit show about the UCI item-score sum and class target?
- RQ3: What are exploratory LR/RF results on the separate Indian modified INDT-ASD cohort?
- RQ4: What local/global model behavior does TreeSHAP expose, and what cannot be concluded?
- RQ5: How can such outputs be communicated to caregivers without presenting them as clinical diagnosis/probability?

12. Paper-ready result narrative and reporting checklist

Draftable factual result narrative

“After removing two exact duplicates, the UCI dataset contained 290 records and was divided into 217 training and 73 test records using a stratified 75/25 split (random seed 17). On the test set, Logistic Regression achieved accuracy 0.9726, balanced accuracy 0.9737, F1 0.9722, ROC AUC 1.0000, and Brier score 0.0192. Random Forest achieved 0.9452, 0.9440, 0.9412, 0.9887, and 0.0945, respectively. XGBoost achieved 0.9726, 0.9726, 0.9714, 0.9977, and 0.0375. The deterministic rule `sum(A1-A10) >= 7` achieved accuracy, balanced accuracy, and F1 of 1.0. Since the UCI class is defined by this rule, these model metrics evaluate reproduction of a score-derived label and do not establish clinical diagnostic performance.”

“In the separate Indian cohort analysis, one 180-month record outside the study article’s stated age range was excluded, leaving 224 records (127 ASD and 97 non-ASD), with 168 training and 56 test records. Logistic Regression achieved accuracy 0.8571, balanced accuracy 0.8490, F1 0.8788, ROC AUC 0.9453, and Brier 0.0876; Random Forest achieved 0.8929, 0.8802, 0.9118, 0.9674, and 0.0731. These exploratory results are based on a single small holdout from a tertiary referral cohort and do not represent population screening performance.”

Required wording rules

- Say “UCI source-data class” or “questionnaire-derived label”, not “clinical diagnosis”.
- Label the Indian S2 study separately. Note that the source cohort is referral-selected.
- Give sample sizes next to percentages and report the one-split design.
- Do not claim significance or model superiority from point metrics alone.
- Name model for every SHAP figure; do not mix RF global explanation with primary XGB local API output.
- Do not call probability calibrated or the P-tiers validated confidence.
- Explain the current UI fixed/default demographic values and question wording adaptation.
- State ethics/privacy/clinical status accurately; do not imply approval or deployment absent evidence.

Recommended figure/table list

- Table 1: cohort characteristics and target mapping.
- Table 2: UCI models plus deterministic rule and demographics-only sensitivity.
- Table 3: independent Indian cohort results.
- Table 4 / appendix: subgroup counts, sensitivity, Wilson intervals; flag small n.
- Figure 1: data-processing/model/application workflow.
- Figure 2: UCI score-to-class cross-tab or target rule diagram.
- Figure 3: cohort EDA with source-specific labels.
- Figure 4: comparative RF global mean absolute SHAP and its exact scope.
- Figure 5: RF local example and/or XGB local example, clearly marked which model and scale.

13. Review/viva Q&A; preparation

What is the main scientific finding?

UCI class is exactly determined by whether A1-A10 sum to at least 7. The models largely reproduce that rule. This is not clinical ASD prediction.

Why include machine learning if the rule is deterministic?

The deterministic baseline and audit reveal what the ML metrics mean. The model comparison characterizes reproduction behavior; it does not show that ML is better clinically.

Is this leakage?

The direct `result` feature is excluded. However, the retained items define the target. Call it a score-derived target/circularity issue; do not say the result field itself was a model input.

Why XGBoost?

The project brief emphasizes tabular structured data and explainability. XGBoost is the UCI primary artifact; it captures nonlinear tree interactions and supports TreeSHAP. We did not show it outperforms LR significantly.

Why does the paper include Indian data?

It is a separate analysis with a different diagnostic reference target. It gives exploratory context but is not a UCI validation cohort or deployed model.

What exactly does SHAP explain?

Attribution of model output among input features relative to a baseline. It explains model behavior, not causal or clinical reasoning.

What is your SHAP validation?

For one held-out comparative RF example, baseline plus contributions reconstructed positive-class probability with absolute error 1.89e-15. XGBoost additivity and broader explanation quality were not verified.

Is your output probability clinically meaningful?

No. It is a probability for the UCI dataset class, whose rule is based on the same ten answers. Calibration and clinical meaning have not been established.

Do you have a deferral system?

The API marks P2/P3 as deferred using uncalibrated thresholds. The current UI still displays the class label, so end-to-end abstention is incomplete.

What fairness analysis did you do?

Descriptive subgroup counts, accuracy/sensitivity, and Wilson intervals: UCI by gender/ethnicity and Indian by gender. Small cells and single split do not support fairness guarantees.

What did you contribute?

Dataset and target audit, comparative exploratory modeling, explicit UCI circularity finding, limited SHAP analysis, and an explainable caregiver-facing research prototype.

What remains?

Repeated/nested and external validation, calibration and decision analysis, true UI abstention, XAI fidelity/stability/usability review, stronger subgroup data/analysis, and clinical/privacy governance. Video and counterfactuals are future scope.

14. Instructions for teammates using an LLM

Supply this PDF plus the project source files identified below. Use the following instruction as the first prompt, then ask the LLM for one paper section at a time.

PROJECT WRITING GUARDRAILS

You are helping draft an academic paper about AutInsight. Treat the supplied handover and repository files as the source of truth. Do not invent experiments, sample sizes, p-values, clinical validity, regulatory/ethics approvals, deployment, calibration, fairness, counterfactuals, or explanation validation. Distinguish UCI score-derived target results from the separate Indian DSM-5-target cohort. Never describe UCI metrics as ASD diagnostic accuracy. Identify which model produced each result or SHAP value. State that evaluation used a single stratified 75/25 split with seed 17 and no external validation. Treat the caregiver questionnaire as adapted and not validated. If source files conflict, report the discrepancy and ask the project team to resolve it; do not silently choose. Use citations for all external/data claims and only include references that can be verified.

Recommended workflow

- Ask for a section outline first, then draft Methods and Results from code/JSON before writing claims in Introduction/Discussion.
- For every numeric claim, request the exact source path and recompute it from the experiment result JSON.
- Ask the model to maintain separate UCI and Indian tables and to label cohort, n, split, model, and target in each.
- Ask for a claim audit: highlight any sentence that implies diagnosis, clinical performance, generalization, fairness, calibration, or validated uncertainty without evidence.
- Have a human verify all dataset and literature references and ensure the university format/citation style is followed.

Never feed raw identifiable health records to an external LLM. Raw data is not needed to draft from the reported artifacts.

15. Project source map, provenance, and references

Question	Start with these files
What did the model actually train?	`scripts/train_models.py`; `scripts/train_indian_models.py`; `models/artifact_manifest.json`.
What were the exact results?	`experiments/results/model_metrics.json`; `experiments/results/indian_model_metrics.json`.
What was audited?	`scripts/validate_data.py`; `scripts/validate_indian_data.py`; `experiments/reports/data_quality.json`; `experiments/reports/indian_data_quality.json`; `experiments/reports/eda_summary.json`; `experiments/reports/indian_eda_summary.json`.
What does the API do?	`backend/app/main.py`; `backend/app/schemas/prediction.py`; `backend/app/services/runtime.py`; `backend/app/core/config.py`.
What does the user see?	`frontend/src/app/page.tsx`; `frontend/src/app/screening/page.tsx`; `frontend/src/lib/questionnaire.ts`; `frontend/src/app/results/page.tsx`; `frontend/src/app/results/understand/page.tsx`.
What are limits and statuses?	`docs/methodology.md`; `docs/model_card.md`; `docs/limitations.md`; `docs/requirements_traceability.md`; `docs/threat_model.md`; `docs/adr/`.
How can environment be reproduced?	`requirements.txt`; `pyproject.toml`; `frontend/package.json`; `docs/reproducibility.md`; `docs/TEAMMATE_SETUP.md`.
What was originally proposed?	`/Users/aadvika/Downloads/Phase 1 Review II ASD (2).pptx` (slides describe intent and literature; not evidence that planned work was implemented).

Current verification recorded by project docs

Backend API suite: four tests were recorded as passing for readiness/metadata, invalid input, genuine inference/SHAP output, and metrics endpoints. Ruff passed. Frontend TypeScript and ESLint passed after caregiver-flow changes. The project documentation says production Next build was run before the latest caregiver-flow revision; no Playwright/E2E suite exists. CI workflow exists but hosted CI was not run. No remote deployment was performed.

Primary references to verify and cite

UCI Machine Learning Repository, Autism Spectrum Disorder Screening Data for Children, dataset 419, DOI 10.24432/C5659W. https://archive.ics.uci.edu/dataset/419/autistic+spectrum+disorder+screening+data+for+children

PLOS ONE, 2019, “Development and validation of DSM-5 based diagnostic tool for children with Autism Spectrum Disorder,” DOI 10.1371/journal.pone.0213242. https://doi.org/10.1371/journal.pone.0213242

Lundberg, S. M., and Lee, S.-I. (2017). “A Unified Approach to Interpreting Model Predictions.” NeurIPS. https://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions

SHAP documentation: https://shap.readthedocs.io/

AQ-10 Child Version source guide used for source item-order/scoring alignment: https://northdevoncoastalpcn.nhs.uk/media/hvpg30ya/aq-10child-autism.pdf

The review presentation also lists several papers and numerical findings from the literature. Verify those items against their original publications before citing them. Slide 12’s neural-model metrics and JSD/entropy coverage values are not present in this repository’s experiment outputs and must not be attributed to these implemented experiments.

End of handover. Keep the core distinction clear: this is an explainable research prototype built around datasets with different label meanings; it is not a clinically validated screening product.